

type of evolving state as width grows

operators
 $n \times n$ matrices

populations
 n -vectors, finitely many

width-free
scalars and kernels

Dense network
 Hn^2 , Markovian

Response memory
 $HmnP$ moving, W_0 fixed

exact scalar closure fails
restricted class, §5
NTH, order p : m^p
proved near the lazy limit

eliminate
learned weights

compress time
 K steps $\rightarrow P$ moments

Stored response history
 $HmnK$, exact

$n \rightarrow \infty$

Tensor Programs
and DMFT
 Hm^2K^2 , exact limit

off the axis, each exact about a different object:

neural tangent kernel (frozen features) · deep linear (no nonlinearity) · one hidden layer (no depth)

grows with steps fixed in time